

B-10-Z

Roll No.....

Total No. of Questions : 16]

[Total No. of Printed Pages : 4

12thARF(SZ)JKUT2024-25

310-Z

BOTANY

(Biology)

Time : 1½ Hours]

[Maximum Marks : 35

SECTION-A

1 each

1. Which of the following is wrongly matched ?

- | | | |
|-------------------|---|--------------------|
| (A) AUG, GUG | – | Methionine, Valine |
| (B) UAA, UAG, UGA | – | Ochre, Amber, Opal |
| (C) AUG, GUA | – | Initiation Codons |
| (D) UAA, UAG, UGA | – | Stop Codons |

2. Bt-toxin kills :

- | | |
|---------------|-------------------|
| ✓ (A) Insects | (B) Weeds |
| (C) Rodents | (D) None of these |

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Turn Over

3. Biogas production is carried out by :
- (A) Halophiles
 - (B) Thermoacidophiles
 - (C) Luminescent bacteria
 - ✓(D) Methanogens
4. The flow of energy in an ecosystem is :
- ✓(A) Unidirectional
 - (B) Bidirectional
 - (C) Multidirectional
 - (D) No specific direction
5. Which *one* of the following is not a hotspot of biodiversity in India ?
- (A) Eastern Ghats
 - (B) Western Ghats
 - (C) Indo-Burma Region
 - (D) Sundaland

SECTION-B

2 each

6. Define Polyembryony. Differentiate between simple and adventive polyembryony.
 7. What is Cellular Totipotency ?
 8. Differentiate between Mutualism and Competition.
 9. Define Pyramid of energy. Explain why pyramid of energy is always upright ?
 10. Name few outbreeding devices. Briefly describe any *one* of them.
- <https://www.jkboseonline.com>

SECTION-C

3 each

11. How does a microspore develop into two male gametes in angiosperms ? Describe the process with suitable diagrams.
12. Define Pleiotropic Gene. Give examples, one each from plants and animals.
13. Describe various applications of Plant Tissue Culture.

.14. Briefly describe the following processes involved in decomposition :

(i) Fragmentation

(ii) Catabolism

(iii) Leaching

15. What are Ramsar Sites ? What is their importance ? Name some Ramsar Sites of India.

SECTION-D

5

.16. What is an Operon ? Explain an Inducible Operon.

Or

State the principle of Independent Assortment. Explain it with a dihybrid cross. Name *one* exception to this principle.